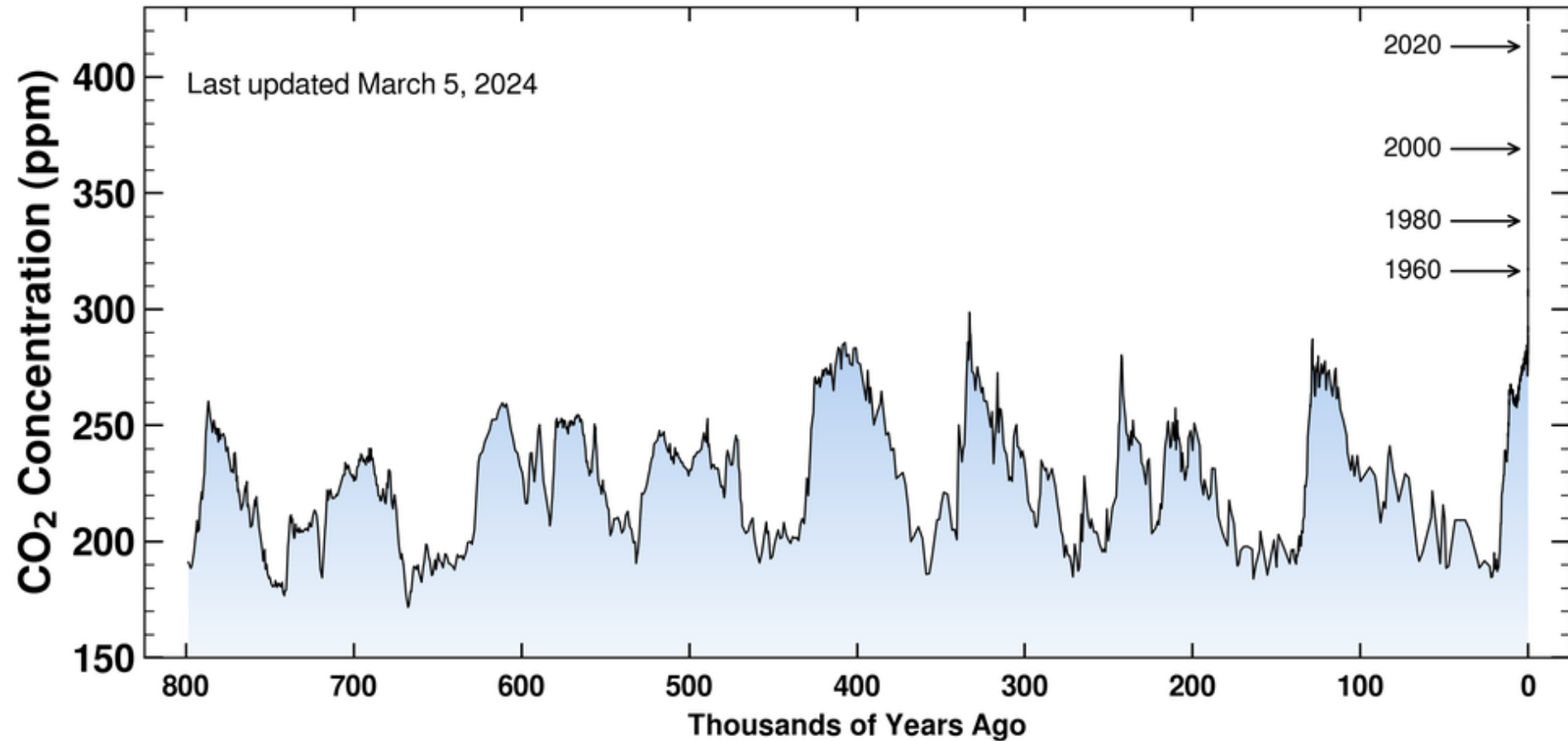


Lecture 11 – Time Series I



Today's Learning Outcomes

1. Know how to describe a periodic signal mathematically (i.e. period, amplitude, etc.)
2. Be able to explain how sampling interval and period of a signal can interact to create aliasing
3. Be able to explain the logic behind the turning points and phase length tests of signal
4. Be able to explain the logic behind the number of point increase and Kendall's Tau approaches to test for trends in a time series

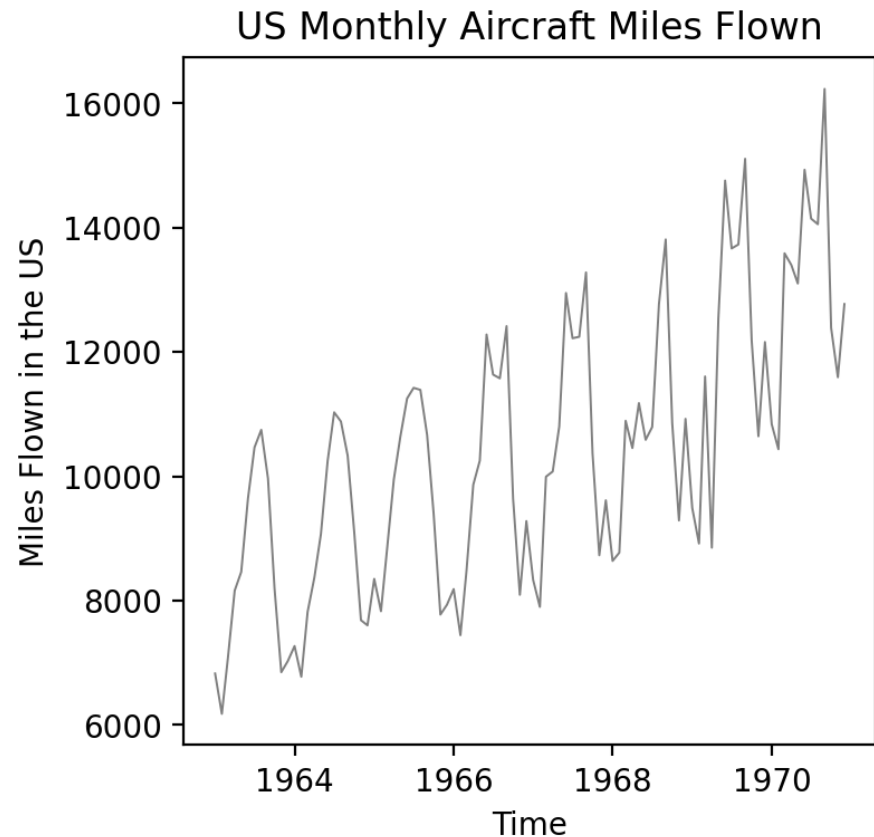
Objectives of Time Series Analysis

1. Develop a relatively simple (i.e. parsimonious) mathematical description of a phenomena (a summative or exploratory approach)
2. Try to explain the behavior of a system in terms of predictor variables or factors (i.e. hypothesis testing)
3. Forecasting of future events, based on either empirical approaches or model driven methods
4. Using objective 1, 2, & 3 to control future events, by intervening in some way
5. Trying to understand the relationship between multiple time series (i.e. the correlation of time series)

Example: Monthly miles traveled by UK airliners (1963-1970)

Has two features
common to time series

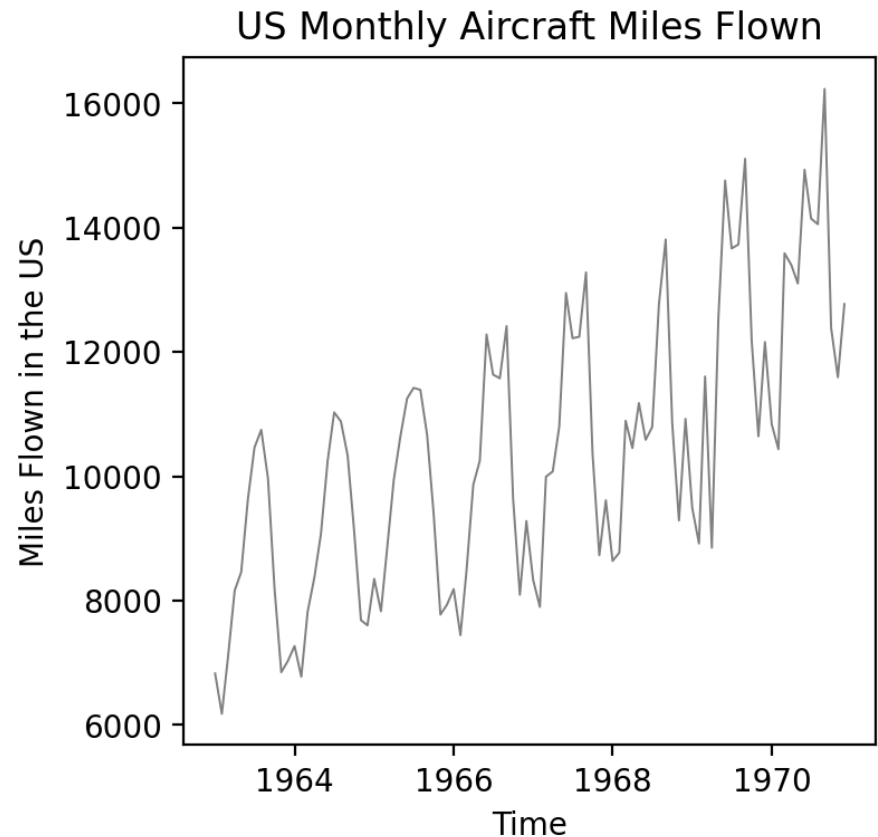
- Yearly “periodicity”
- General upward trend



Breaking Down Time Series

Ideally, we want to break down our time series into component of...

1. Trend(s)
2. Periodic signal(s)
3. Noise/Error



Parts of Time Series to Model

We can think of a time series as being composed of a 3 parts or categories

1. A **trend** or long-term movement

- Often linear, but does not have to be

A single time series might have multiple signals within each of these categories

2. **Periodic fluctuations** around the mean or trend

- May include detectable aperiodic events as well

3. **Noise**, the residual, irregular or random effect

Core question for time series analysis

**Do we see signal (trend/
periodicity/structure), or only
noise?**

What it means:

Before modeling signal, we must test the “pure noise” hypothesis.

What “pure noise” means for ordered data

Null hypothesis (H_0)

- The observations are *i.i.d.* with a continuous distribution; the time order carries no information, i.e., the values are equally likely to appear in any order.

What H_0 rules out

- No trend (level doesn't drift)
- No seasonality/periodicity (no systematic cycles)
- No autocorrelation/persistence (today doesn't predict tomorrow)

Why the order matters

- In time series, time is directional. If shuffling the data doesn't change its properties, then the ordering adds nothing—that's pure noise.

Statistical Tests for Detecting Structure in a Time Series

1. Tests for Detecting Any Structure (Deviation from Pure Noise)

These tests check whether the data's ordering contains information, i.e., whether the series is more smooth or more erratic than expected under randomness.

1.1. Turning Points Test

Detects whether the series changes direction too often or too rarely.

1.2. Phase-Length Distribution Test

Compares observed lengths of monotonic segments to what would be expected if the order were random.

Statistical Tests for Detecting Structure in a Time Series

2. Tests for Detecting a Trend

These tests look for a systematic upward or downward pattern.

2.1. Number of Increases / Sign Test for Trend

Checks whether upward steps occur more often than downward steps.

2.2. Kendall's Tau (Rank Correlation)

A nonparametric test that measures monotonic trend in the ordered data.

3. Tests for Detecting Periodicity

Tests focused on detecting repeating seasonal or cyclical patterns.

3.1. Friedman's Test for Seasonality

A rank-based test that identifies whether values repeat consistently across time blocks (seasons).

1.1. Turning Points Test

A **turning point** in a numerical sequence $x_1, x_2, \dots, x_n$ is a point x_i where the sequence changes direction:

A **peak**:

$$x_{i-1} < x_i > x_{i+1}$$

A **trough**:

$$x_{i-1} > x_i < x_{i+1}$$

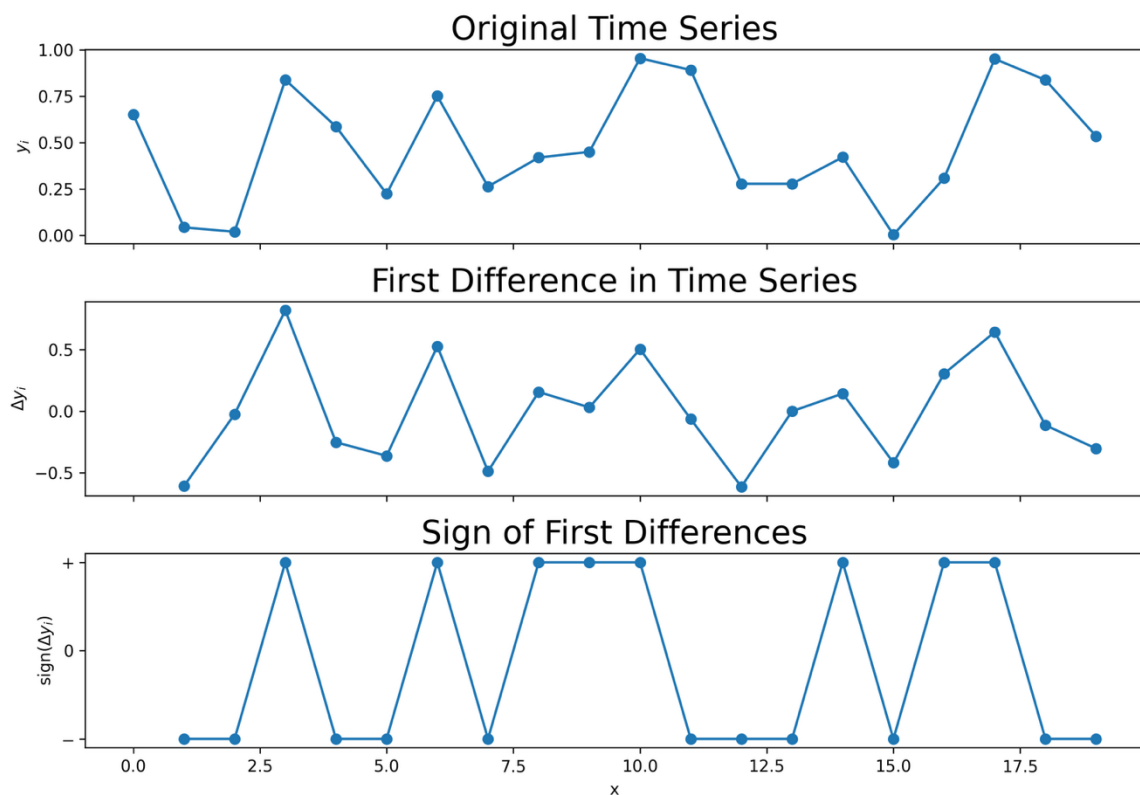
The total number of turning points is denoted by p .

Turning point counts are used as a test of **randomness**:

- A purely random sequence fluctuates frequently → many turning points
- A smooth or autocorrelated sequence has fewer turning points

This forms the Turning Point Test for independence.

Turning Points Example



Turning Points-
Where the curve
changes direction

First Differences
 $\text{diff}(x,1)$ -
differences
between
consecutive
points

Every time the sign of
the first difference
changes, we have a
turning point

Easy way to count
turning points

n = number of time steps

Variance and z-scores

For an *i.i.d.* continuous random sequence (i.e., all permutations equally likely):

$$E[p] = \frac{2n - 4}{3}$$

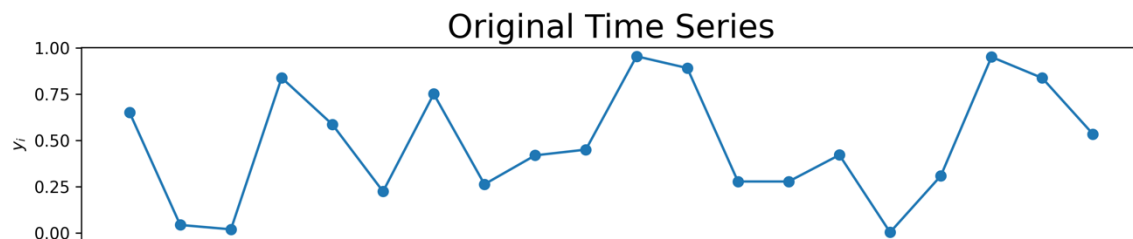
$$variance(p) = \frac{16n - 29}{90}$$

and the Z-test statistics is

$$z = \frac{p - \frac{2n - 4}{3}}{\sqrt{variance(p)}}$$

- Large $|z| \rightarrow$ too many or too few turning points $\rightarrow$ reject randomness
- Small $|z| \rightarrow$ consistent with a random sequence

Turning Points Test Example



```
from turning_point import turning_point_test
```

```
x = np.array(np.random.default_rng(43).random(20),  
dtype=float)
```

```
# make sure that the input data into  
turning_point_test is a list, not a numpy array  
turning_point_test(x.tolist())
```

Output

```
TurningPointTestResult(n=20, n_effective=20, T=10,  
mu=12.0, var=3.2333333333333334, z=-  
0.8341919974800728, p_value=0.404172799917852)
```

1.2. Phase-Length Distribution Test

Definition of a Phase

A **phase** is a sequence of consecutive steps that all move in the **same direction**, either increasing or decreasing.

Example:

$3 \rightarrow 5 \rightarrow 7$ is an **increasing phase**

$7 \rightarrow 6 \rightarrow 4$ is a **decreasing phase**

Phases help identify how smooth or noisy a sequence is.

Extracting Phase Lengths

To analyze a dataset, we extract the lengths of these phases:

- In **R**, the function `rle()` (Run-Length Encoding) automatically identifies consecutive runs and their lengths.
- **Python** does not include `rle()` natively, so we implement a custom function to produce the same output.

This produces a list of phase lengths such as:

`[1, 3, 2, 4, 1, ...]`

representing how long each monotonic segment lasts.

Phase-Length Distribution Test

A nonparametric test that compares the *observed* distribution of monotonic phase lengths in a time series to the *expected* distribution under complete randomness (*i.i.d.*, no trend, no persistence).

Test statistic:

$$X^2 = \sum_d \frac{\left(N_d^{obs} - E(N_d)\right)^2}{E(N_d)}$$

This is a classic *chi-square* goodness-of-fit test.

- Large value → distribution differs from random permutation
- Small value → compatible with randomness (no signal)

For large n:

$$X^2 \sim \chi_k^2$$

where k = number of non-empty phase-length categories minus 1.

Analytical Expectation Under a Random Ordering

If the data behave like a **uniform random permutation** (i.e., all orderings are equally likely), the *expected* number of phases of each length can be computed analytically.

For a sequence of length n , the expected number of phases of length d (N_d) is:

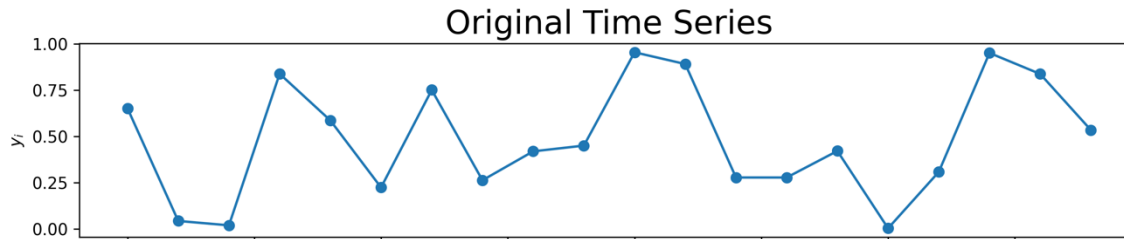
$$E(N_d) = \frac{2(n-d-2)(d^2+3d+1)}{(d+3)!}$$

valid for:

$$1 \leq d \leq n-3$$

This formula comes from combinatorial theory of monotone runs in random permutations.

Phase Length Distribution Test Example



```
from phase_length_test import  
phase_length_distribution_test,monotonic_phase_lengths  
monotonic_phase_lengths(x)
```

Output

```
[3, 2, 3, 2, 2, 4, 4, 2, 2, 3, 3]
```

```
# apply the phase length distribution test to the  
data
```

```
res = phase_length_distribution_test(x, B=2000,  
seed=1)
```

```
print(res.p_perm, res.statistic)
```

Output

```
0.17841079460269865 2.2272727272727266
```



We talked about
detecting
randomness of the
time series.

Next, we move on
to testing trend!

2.1. Testing for Trend: Counting Increases

Procedure

Count the number of times the series increases:

$$c = \#\{x_{i+1} > x_i\}$$

Under the **null hypothesis** (*i.i.d.*, no trend):

- increases and decreases are equally likely

$$P(\text{increase}) = P(\text{decrease}) = \frac{1}{2}$$

Therefore, the number of increases behaves like a Binomial random variable with $n - 1$ trials.

Expected Behavior Under the Null Model

Expected mean value is:

$$E(c) = \frac{1}{2}(n - 1)$$

The variance is

$$\text{Var}(c) = \frac{1}{2}(n + 1)$$

The variance uses a classical small-sample correction from time-series statistics.)

Trend Test Statistic and Interpretation

Z-test Statistic

$$Z = \frac{c - E(c)}{\sqrt{\text{Var}(c)}}$$

For large n ,

$$Z \sim N(0,1)$$

Interpretation

- $Z > 0 \rightarrow$ more increases than expected $\rightarrow$ evidence for **upward trend**
- $Z < 0 \rightarrow$ fewer increases $\rightarrow$ evidence for **downward trend**
- **small $|Z|$** $\rightarrow$ consistent with **random noise** (no trend)

Alternative Approach: Permutation Test

Instead of relying on theory:

- Shuffle the data many times
- Compute the number of increases for each shuffle
- Compare the observed value to this empirical distribution

This gives an exact, assumption-free trend test.

Counting Increases Example

```
from count_increase_test import count_increases_test
```

```
# Example 1: no trend (i.i.d. noise)
```

```
import numpy as np
```

```
rng = np.random.default_rng(0)
```

```
x = rng.normal(size=60)
```

```
res = count_increases_test(x, alternative="two-sided")
```

```
print(res)
```

```
# Example 2: increasing trend
```

```
t = np.arange(60)
```

```
x_trend = 1 * t + rng.normal(scale=0.5, size=60)
```

```
res_inc = count_increases_test(x_trend,  
                               alternative="increasing")
```

```
print(res_inc)
```

```
# Output
```

```
CountIncreasesResult(n=60, n_trials=59, n_increases=32,  
n_decreases=27, n_ties=0, alternative='two-sided', p_value=0.603)
```

```
CountIncreasesResult(n=60, n_trials=59, n_increases=52,  
n_decreases=7, n_ties=0, alternative='increasing',  
p_value=6.794959e-10)
```

2.2. Testing for Trend: Kendall's Tau

What Kendall's Tau Measures

- Kendall's Tau is a **non-parametric measure of monotonic association** between a variable x and time t .
- Positive Tau $\rightarrow$ values generally increase over time (upward trend)
- Negative Tau $\rightarrow$ values generally decrease over time (downward trend)

Definition

- Tau is computed from the number of **discordant pairs**, Q :

$$\tau = 1 - \frac{4Q}{n(n-1)}$$

- A *discordant pair* is a pair (x_i, x_j) where the ordering in time disagrees with the ordering in value.
- Fewer discordant pairs $\rightarrow$ stronger trend.

Kendall's Tau: Inference and Testing

Sampling Distribution Under the Null (No Trend)

$$\text{Var}(\tau) = \frac{2(2n + 5)}{9n(n - 1)}$$

Z-test Statistic

$$Z = \frac{\tau}{\sqrt{\text{Var}(\tau)}}$$

For large n :

$$Z \sim N(0,1)$$

You can compare this Z-value to the standard normal distribution (z-table).

Kendall's Tau example

```
t = np.arange(60)
rng = np.random.default_rng(0)
x_trend = 1 * t + rng.normal(scale=0.5, size=60)
tau, p_value = kendalltau(t, x_trend)
print("Kendall tau:", tau)
print("p-value:", p_value)
```

#Output

```
Kendall tau: 0.9966101694915255 p-value:
2.2985232984630205e-29
```



Lastly, we move on
to testing
periodicity!

Testing for Periodicity: Conceptual Background

How periodicity shows up in a time series

- A periodic system is often symmetrical, with repeated ups and downs.
- Over time, every “up” is countered by a later “down.”

Implications for earlier tests

- A strongly periodic time series tends to show:
 - More turning points than expected under randomness
 - Fewer long monotonic phases (because the series reverses direction regularly)
- But: these require long time series

Turning-point tests and phase-length tests need **large n** before they reliably detect periodic structure.

Limitations of Turning-Point and Phase Tests for Periodicity

Why these tests *struggle*? We want to detect the alternative hypotheses:

- H_1 : Trend + noise
- H_2 : Pure noise (i.i.d.)
- Periodic signal: ups and downs at consistent intervals

But turning-point and phase tests often cannot distinguish:

- periodicity from trend or autocorrelated noise

unless the time series is very long.

Motivation

Because periodicity is a structured pattern in *time blocks*, we need a test that directly examines repeatability across cycles.

This leads to **Friedman's Test**.

3.1. Friedman's Test for Seasonality (Annual Example)

Friedman (1937) proposed a non-parametric ANOVA-type test

Used to detect **seasonality or periodicity** in repeated time blocks (e.g., months within years).

Data preparation

1. (Optional but recommended) Remove any linear trend.

2. For each year:

- Rank the 12 monthly values from 1 to 12.

3. Construct a matrix:

- Rows (r): months (seasons)
- Columns (c): years

4. For each month j , sum its ranks across years $\rightarrow M_j$

This creates a seasonality score for each month.

Friedman's Test of Seasonality: Hypothesis & Test Statistic

H_0 : No seasonality

- Monthly patterns do **not** align across years.
- Therefore, the rank sums M_j should be **approximately equal** across months.

H_1 : Seasonality exists

- Certain months consistently rank high or low every year.

Friedman's test statistic T approximates a **Chi-Square distribution** under H_0 , with degree of freedom ($df = r - 1$):

$$T = \frac{12}{c r(r + 1)} \sum_{j=1}^r M_j^2 - 3c(r + 1)$$

where:

- r = number of seasons (e.g., 12 months)
- c = number of years (repeated blocks)

A large T indicates significant seasonality.

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